

PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn

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Abstract—Modern Wi-Fi deployments increasingly suffer from spectrum underutilization as overlapping Basic Service Sets (OBSSs) compete for primary channel access. IEEE 802.11bn introduces Non-Primary Channel Access (NPCA) to exploit these underutilized intervals, yet the standard contention mechanism is fundamentally mismatched to NPCA’s constrained transmission windows: backoff counters routinely exceed the remaining opportunity duration, wasting slots that could otherwise carry data. We propose PACE, a probabilistic adaptive contention control scheme that enables stations to collectively converge toward a theoretically optimal transmission probability without central coordination. PACE combines a distributed multiplicative feedback rule—copying the successful sender’s probability on solo receptions—with a prospective self-exclusion mechanism that silences stations whose frame length exceeds the remaining window. We evaluate PACE through simulation spanning a wide range of station counts and window lengths, comparing against Binary Exponential Backoff and several competing adaptive schemes under both homogeneous and heterogeneous frame-length distributions. PACE consistently outperforms standard backoff in channel utilization and achieves Pareto-dominance in the throughput–fairness tradeoff. The advantage is most pronounced when the available window is scarce relative to the contending population, precisely the regime in which standard backoff wastes the largest share of transmission opportunities. These results identify the tight-window regime as the critical operating condition for NPCA and establish PACE as a practical design reference for adaptive contention policy in next-generation Wi-Fi networks.

Index Terms—Non-Primary Channel Access, IEEE 802.11bn, Probabilistic Access Control, Mean Field Game, OBSS, Adaptive Contention, Wi-Fi

I. INTRODUCTION

Spectrum efficiency in next-generation Wi-Fi networks depends critically on how effectively stations exploit available channel time. In dense deployments, overlapping Basic Service Sets (OBSSs) routinely occupy the primary channel while adjacent channels remain idle, creating a persistent underutilization problem. IEEE 802.11bn addresses this gap by introducing Non-Primary Channel Access (NPCA), which allows stations to transmit opportunistically on non-primary channels during ongoing OBSS intervals [1], [2].

NPCA, however, presents a contention challenge distinct from conventional channel access. Each NPCA opportunity has a finite duration bounded by the ongoing OBSS transmission, and competing stations must coordinate their access within this constrained window. The Binary Exponential

Backoff (BEB) mechanism was designed for open-ended contention: each collision doubles the backoff window, and the resulting delay frequently exceeds the remaining NPCA window duration. Affected stations are effectively silenced for the rest of the interval, leaving the channel idle even when other stations could have transmitted. In heterogeneous deployments where stations carry frames of varying lengths, the problem is further compounded: stations whose frame cannot fit in the remaining window continue to occupy contention slots despite having no viable transmission opportunity.

Consider a dense office environment where multiple stations detect an OBSS transmission and attempt to use the secondary channel concurrently. Early collisions trigger exponential backoff growth; within a few rounds, a large fraction of stations have accumulated backoff counts that exceed the remaining window duration. The channel spends the final slots idle, not because no station wanted to transmit, but because the contention mechanism pushed every backoff counter beyond the opportunity horizon.

Existing work on NPCA has focused on characterizing aggregate throughput and delay under the standard switching rule, taking Binary Exponential Backoff within the non-primary channel as given [2]–[4]. The contention dynamics within the finite NPCA window—and the mismatch between BEB’s infinite-horizon design and the window’s bounded duration—have not been addressed. Adaptive probabilistic schemes designed for infinite-horizon neighbor discovery, such as the ALOHA-like Neighbor Discovery (AND) algorithm [5], reduce overhead by avoiding explicit coordination, but their stationary-population assumption makes them ill-suited to a window that depletes over time. BEB’s infinite-horizon design and AND’s open-loop phase structure both fail to track the time-varying optimal access probability as the window shrinks and the set of viable stations changes.

In this paper, we propose PACE, a Probabilistic Adaptive Contention control scheme for NPCA that enables stations to collectively converge toward the theoretically optimal transmission probability for each moment within the window, without central coordination or explicit inter-station signaling. PACE adapts the MIMD feedback rule of the Probabilistic Neighbor Discovery (PND) algorithm [6] to the finite-window NPCA context: stations copy the successful sender’s probability on solo receptions, reduce on collision, and increase on idle

slots, causing distributed dynamics to track the Mean Field Game (MFG)-optimal policy. PACE further adds a PPDU-aware self-exclusion rule that silences stations whose frame length exceeds the remaining window, eliminating a class of futile contention attempts that BEB would otherwise sustain.

The main contributions of this paper are as follows:

- We derive the MFG-optimal transmission probability for finite-window NPCA, showing that the target probability must track both the remaining window duration and the number of still-viable competing stations.
- We propose PACE, a fully distributed algorithm that approximates the MFG-optimal policy through MIMD feedback and PPDU-aware self-exclusion, without requiring any inter-station signaling or knowledge of the total station count.
- Through extensive simulation under homogeneous and heterogeneous frame-length distributions, we demonstrate that PACE achieves significantly higher window utilization than BEB, is Pareto-dominant in the throughput-fairness tradeoff, and maintains proportional fairness between native and visiting stations.

The remainder of this paper is organized as follows. Section II reviews related work on NPCA, adaptive MAC protocols, and mean field game analysis. Section III presents the system model. Section IV describes the PACE algorithm and its theoretical foundation. Section V presents simulation results, and Section VI concludes the paper.

II. RELATED WORKS

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III. SYSTEM MODEL

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IV. PACE ALGORITHM

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V. PERFORMANCE EVALUATION

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VI. CONCLUSION

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